Answer All Three Questions

The numbers in square brackets show the provisional allocation of maximum marks per question or part of question.

A module materials booklet, including physical constants if applicable, is attached at the end of the paper.

[Part marks]

1. (a) Consider the following states:

$$|\psi_1\rangle = |\uparrow\rangle \otimes |\uparrow\rangle \qquad |\psi_2\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle\right)$$

$$|\psi_3\rangle = |\downarrow\rangle \otimes |\downarrow\rangle \qquad |\psi_4\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\rangle \otimes |\downarrow\rangle + |\downarrow\rangle \otimes |\uparrow\rangle\right)$$

Which of these states are:

i. symmetric? **[1]**

ii. anti-symmetric? **[1]**

iii. a product state (i.e. not entangled)? **[1]**

(b) A Helium atom is in its ground state and its two electrons are both occupying the same spatial state. Which of the 4 states above would be a valid spin state for these electons, and why? **[3]**

(c) Using a column vector representation for the basis states $|\uparrow\rangle, |\downarrow\rangle$

$$|\uparrow\rangle \rightarrow \begin{bmatrix}1\\0\end{bmatrix} \qquad |\downarrow\rangle \rightarrow \begin{bmatrix}0\\1\end{bmatrix},$$

show that these states are transformed by Pauli operators σ_x and σ_z as follows:

$$\sigma_z|\uparrow\rangle = |\uparrow\rangle \qquad \sigma_z|\downarrow\rangle = -|\downarrow\rangle \qquad \sigma_x|\uparrow\rangle = |\downarrow\rangle \qquad \sigma_x|\downarrow\rangle = |\uparrow\rangle$$

[4]

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(d) A pair of coupled spin-half particles has the following Hamiltonian:

$$\hat{H} = -\Delta(\sigma_x \otimes \sigma_x + \sigma_z \otimes \sigma_z)$$

where $\Delta > 0$.

Using the variational method with the parametrised Ansatz state $|\psi(a)\rangle = |\uparrow\rangle \otimes |\uparrow\rangle + a|\downarrow\rangle \otimes |\downarrow\rangle$, where a is a real parameter, provide an estimate and upper bound to the ground state energy of $\hat{H}$. In your answer, include the value of a for which the energy of $|\psi(a)\rangle$ is minimized. **[8]**

(e) Imagine that the sign of the parameter Δ changes, so that Δ is replaced by $-\Delta$ in the Hamiltonian $\hat{H}$. What upper bound to the ground state energy does your variational model now predict for this new Hamiltonian, and how might the prediction be improved? **[2]**

2. The total angular momentum operator $\hat{J} = (\hat{J}_x, \hat{J}_y, \hat{J}_z)$ has associated ladder operators $\hat{J}_+$ and $\hat{J}_-$ for a generalised system.

(a) Derive the equation below, that shows the action of the lowering operator on the state $|j\, m\rangle$:

$$\hat{J}_+|j\, m\rangle = \sqrt{j(j+1) - m(m+1)}\, \hbar|j\, m+1\rangle$$

[3]

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For the remainder of this question, consider a system of two spin-$\frac{1}{2}$ particles labelled 1 and 2. The total spin operator is $\hat{\mathbf{S}} = \hat{\mathbf{S}}_1 + \hat{\mathbf{S}}_2$, and the Hamiltonian for the system is given by:

$$\hat{H} = \Gamma\,(\hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_2) + \lambda\,\hat{S}_z^2,$$

where Γ and λ are real constants.

(b) Show that the total spin operator $\hat{\mathbf{S}}^2$ can be expressed in terms of $\hat{\mathbf{S}}_1^2$, $\hat{\mathbf{S}}_2^2$, and $\hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_2$. Use this to rewrite the Hamiltonian in terms of $\hat{\mathbf{S}}^2$ and $\hat{S}_z^2$ only. **[4]**

(c) Compute the following expectation values for the state, $|S = 0, M = 0\rangle$: $\langle \hat{S}_{1z} \rangle$, $\langle \hat{S}_{2z} \rangle$, and $\langle \hat{S}_{1x}\hat{S}_{2x} \rangle$. **[4]**

(d) Form the matrix representation of $\hat{H}$ in the **total spin** basis: $|S = 1, M = 1\rangle, |S = 1, M = 0\rangle, |S = 1, M = -1\rangle$, and $|S = 0, M = 0\rangle$. Clearly explain how the matrix elements are computed including any coupling present. The matrix should be ordered in the following way: **[6]**

$$\begin{pmatrix} (\frac{\Gamma}{4} + \lambda)\hbar^2 & 0 & \cdots \\ 0 & (\frac{\Gamma}{4})\hbar^2 & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

(e) By reordering the columns of your matrix representation, determine the energy eigenvalues of this system. **[3]**

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[Part marks]

3. (a) A one-dimensional quantum harmonic oscillator has the Hamiltonian:

$$\hat{H}_0 = \frac{\hat{p}^2}{2m} + \frac{m\omega^2\hat{x}^2}{2}$$

Re-express $\hat{H}_0$ in terms of raising and lowering operators, explaining your notation. **[3]**

(b) The system is now subject to an additional potential which is a quartic function of x. The Hamiltonian of this system is:

$$\hat{H} = \hat{H}_0 + \lambda\kappa\hat{x}^4 \ ,$$

where $\kappa = m^2\omega^3/\hbar$ and $\lambda > 0$ is a real dimensionless quantity. Show that this Hamiltonian can be written in the form:

$$\hat{H} = \hat{H}_0 + \lambda\gamma(\hat{a}_- + \hat{a}_+)^4 \ ,$$

and identify γ. **[1]**

(c) Show that:

$$(\hat{a}_- + \hat{a}_+)^4|0\rangle = 3|0\rangle + 6\sqrt{2}|2\rangle + \sqrt{4!}|4\rangle \ .$$ **[4]**

(d) Hence, or otherwise, calculate the first-order correction to the energy of the ground state of the system. **[2]**

(e) Compute the second-order correction to the ground state energy, and hence write down an expression for this energy which is correct to second order in λ. **[6]**

(f) Show that for a Hermitian operator $\hat{A}$, $\langle\psi|\hat{A}^4|\psi\rangle = \langle\phi|\phi\rangle$, where $|\phi\rangle = \hat{A}^2|\psi\rangle$. **[1]**

(g) Hence, or otherwise, calculate the first-order correction $E_n^{(1)}$ to the n-th energy eigenvalue. **[3]**

END OF EXAMINATION PAPER